ADVANCED OPTIMISATION

1. a) i) The problem is

$$\begin{aligned} \max_{p_k} \quad & \min_{i=1,\dots,n} \frac{S_i}{I_i+\sigma_i} \\ s.t. \quad & 0 \le p_i \le P_i^{\max}, \quad i=1,\dots,n, \\ & \sum_{k\in K_j} p_k \le P_j^{gp}, \quad j=1,\dots,m, \\ & \sum_{k=1}^{n} G_{ik}p_k \le P_i^{rc}, \quad i=1,\dots,n. \end{aligned}$$

1 mark to the objective function, 1 mark for each constraint (four constraints). 1 mark extra mark for everything correct [6 marks]

ii) Since the problem is $\max f(x)$, we want to be able to test when $f(x) \ge t$ and push this t as high as possible. Thus we look for a family of convex functions ϕ_t parametrised in t such that

(I) $f(x) \ge t \iff \phi_t(x) \le 0$ for each x.

(II) $\phi_t(x)$ nondecreasing in t.

Condition (II) ensures that if $\phi_t(x) \le 0$ (for all x), then $\phi_s(x) \le 0$ (for all x) for $s \le t$. This ensures consistency as if (I) holds for a high value t, it must hold for all lower values $s \le t$.
We want

$$\min_{i=1,\dots,n} \frac{S_i}{I_i+\sigma_i} \ge t.$$

This is equivalent to

$$\frac{S_i}{I_i+\sigma_i} \ge t, \qquad \text{for all } i=1,\dots,n.$$

From the positivity of $I_i+\sigma_i$

$$t(I_i+\sigma_i) - S_i \le 0, \qquad \text{for all } i=1,\dots,n,$$

follows. So $\phi_t^i(x) = t(I_i+\sigma_i) - S_i$. Note that $s(I_i+\sigma_i) - S_i \le t(I_i+\sigma_i) - S_i$ for any $s \le t$ (again from the positivity of $I_i+\sigma_i$). Thus $\phi_t^i(x)$ satisfies both required properties.

3 marks for the derivation of Φ_t, 1 mark for showing it is nondecreasing. [4 marks]

b) i) The problem, as stated, is

$$\begin{aligned} \max_{T,r,w} \quad & \alpha_4 T r^2 \\ s.t. \quad & \alpha_1 T r w^{-1} + \alpha_2 r + \alpha_3 r w \le C_{\max}, \\ & T_{\min} \le T \le T_{\max}, \\ & r_{\min} \le r \le r_{\max}, \\ & w_{\min} \le w \le w_{\max}, \\ & w \le \frac{1}{10} r. \end{aligned}$$

Answers

This can be written in posynomial form as

$$\begin{aligned}
\min_{T,r,w} \quad & \alpha_4^{-1} T^{-1} r^{-2} \\
s.t. \quad & \frac{\alpha_1}{C_{\max}} T r w^{-1} + \frac{\alpha_2}{C_{\max}} r + \frac{\alpha_3}{C_{\max}} r w \leq 1, \\
& T_{\max}^{-1} T \leq 1, \qquad T_{\min} T^{-1} \leq 1, \\
& r_{\max}^{-1} r \leq 1, \qquad r_{\min} r^{-1} \leq 1, \\
& w_{\max}^{-1} w \leq 1, \qquad w_{\min} w^{-1} \leq 1, \\
& 10 w r^{-1} \leq 1.
\end{aligned}$$

1 mark for the objective function, 1 for the total cost, 1 mark for the three set constraints, 1 mark for the 10% *constraint, 1 mark for writing it in polynomial form.* [5 marks]

ii) Let $y_T = \log T$, $y_r = \log r$ and $y_w = \log w$. The geometric problem in convex form is

$$\begin{aligned}
\min_{y_T, y_r, y_w} \quad & -y_T - 2y_r - \log \alpha_4 \\
s.t. \quad & \log\left(\frac{\alpha_1}{C_{\max}} e^{y_T + y_r - y_w} + \frac{\alpha_2}{C_{\max}} e^{y_r} + \frac{\alpha_3}{C_{\max}} e^{y_r + y_w} \right) \leq 0, \\
& y_T - \log T_{\max} \leq 0, \qquad -y_T + \log T_{\min} \leq 0, \\
& y_r - \log r_{\max} \leq 0, \qquad -y_r + \log r_{\min} \leq 0, \\
& y_w - \log w_{\max} \leq 0, \qquad -y_w + \log w_{\min} \leq 0, \\
& y_w - y_r + \log 10 \leq 0.
\end{aligned}$$

1 mark for the objective function, 1 for the log sum exp, 1 mark for three left set constraints, 1 mark for three right set constraints, 1 mark for the 10% *constraint.* [5 marks]

Answers

2. a) i) Figure 2.1 shows the feasible set (the intersection of the two shaded disks) and some contour lines of the objective function. There is only one feasible point, $x^* = (1,0)$, so it is optimal for the primal problem, and we have $p^* = 1$.

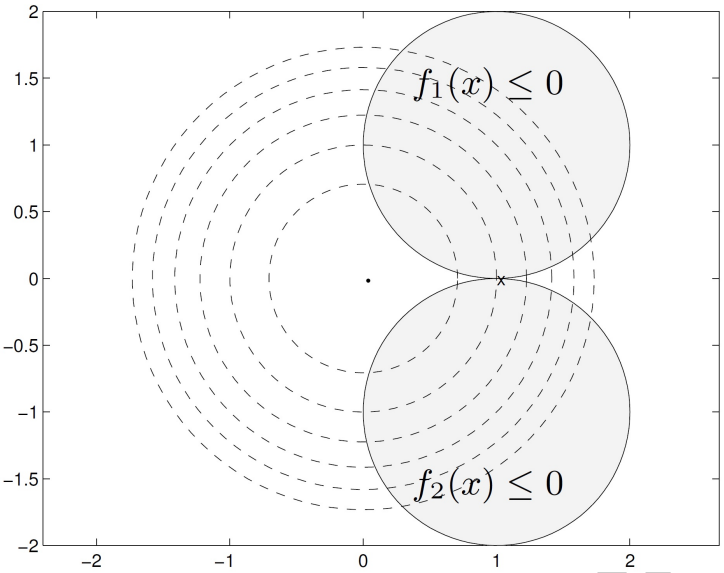


Figure 2.1 Plot for part 2.a).

1 mark for the plot, 1 mark for the optimal point, 1 mark for the optimal value. [3 marks]

ii) The KKT conditions are

$$\begin{aligned} (x_1-1)^2+(x_2-1)^2 &\leq 1, & (x_1-1)^2+(x_2+1)^2 &\leq 1,\\ \lambda_1 &\geq 0, & \lambda_2 &\geq 0,\\ 2x_1+2\lambda_1(x_1-1)+2\lambda_2(x_1-1) &= 0, & 2x_2+2\lambda_1(x_2-1)+2\lambda_2(x_2+1) &= 0,\\ \lambda_1((x_1-1)^2+(x_2-1)^2-1) &= 0, & \lambda_2((x_1-1)^2+(x_2+1)^2-1) &= 0. \end{aligned}$$

At $x^* = (1,0)$, these conditions reduce to

$$\lambda_1 \geq 0, \qquad \lambda_2 \geq 0, \qquad 2 = 0, \qquad \lambda_1 = \lambda_2,$$

which, because of the third condition, do not hold.

1 mark for the KKT conditions, 1 mark for checking the conditions. [2 marks]

iii) The Lagrange dual function is

$$g(\lambda_1,\lambda_2) = \inf_{x_1,x_2} L(x_1,x_2,\lambda_1,\lambda_2).$$

The Lagrangian L is

$$\begin{aligned} &L(x_1,x_2,\lambda_1,\lambda_2)\\ &= x_1^2+x_2^2+\lambda_1((x_1-1)^2+(x_2-1)^2-1)+\lambda_2((x_1-1)^2+(x_2+1)^2-1)\\ &= (\lambda_1+\lambda_2+1)x_1^2+(\lambda_1+\lambda_2+1)x_2^2-2(\lambda_1+\lambda_2)x_1-2(\lambda_1-\lambda_2)x_2+\lambda_1+\lambda_2. \end{aligned}$$

- If $\lambda_1+\lambda_2+1 < 0$, then for $x_1 \to +\infty$ or $x_2 \to +\infty$, we have $g \to -\infty$.

Answers

- If $\lambda_1 + \lambda_2 + 1 = 0$, then

$$L(x_1, x_2, \lambda_1, \lambda_2) = 2x_1 + 4\lambda_2 x_2 + 2x_2 - 1,$$

and for e.g. $x_1 \to -\infty$, we have $g \to -\infty$.

- If $\lambda_1 + \lambda_2 + 1 > 0$, the L reaches its minimum for

$$x_1 = \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + 1}, \qquad x_2 = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + 1}.$$

In summary,

$$g(\lambda_1, \lambda_2) = \begin{cases} \dfrac{\lambda_1 + \lambda_2 - (\lambda_1 - \lambda_2)^2}{\lambda_1 + \lambda_2 + 1}, & \lambda_1 + \lambda_2 + 1 > 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

1 mark for the Lagrangian, 1 mark for each case, 1 mark the dual function. [5 marks]

iv) The Lagrange dual problem is

$$\begin{array}{ll} \max & \dfrac{\lambda_1 + \lambda_2 - (\lambda_1 - \lambda_2)^2}{\lambda_1 + \lambda_2 + 1} \\ s.t. & \lambda_1 \geq 0, \qquad \lambda_2 \geq 0. \end{array}$$

Since g is symmetric, the optimum (if it exists) occurs with $\lambda_1 = \lambda_2$. The dual function then simplifies to

$$g(\lambda) = \frac{2\lambda}{2\lambda + 1}.$$

Computing the derivative, we see that this function is always increasing. Thus, $g(\lambda) \to 1$ as $\lambda \to +\infty$. We have $d^* = p^* = 1$, but the dual optimum is not attained. Despite strong duality holds, the KKT conditions fail because the dual optimum is not attained.

1 mark for the simplified function, 1 mark for optimal value, 1 mark for the justification. [3 marks]

b) i) $||x_{ls}||_2 = 3$.

ii) $||Ax_{ls} - b||_2^2 = 2$.

iii) $||b||_2 = \sqrt{10}$.

iv) About 5.5.

v) About 5.5.

vi) About 6. In fact, from the curve we see that if $||Ax - b||_2^2 = 2$, then $||x||_2^2 = 9$, so 11, if $||Ax - b||_2^2 = 3$, then $||x||_2^2 = 4$, so 7, if $||Ax - b||_2^2 = 4$, then $||x||_2^2 = 2.1$, so 6.1, if $||Ax - b||_2^2 = 5$, then $||x||_2^2 = 1.5$, so 6.5.

vii) The rank of A is 10 since the least square solution is unique.

1 mark for each question. [7 marks]

Answers

3. a) i) We use the second-order condition.

$$f(x) = \log x, \quad f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}.$$

Since $f''(x) < 0$ for all $x \in \mathbb{R}_{++}$, f is concave. [1 mark]

ii) We use the second-order condition.

$$f(x) = x\log x, \quad f'(x) = \log x + 1, \quad f''(x) = \frac{1}{x}.$$

Since $f''(x) > 0$ for all $x \in \mathbb{R}_{++}$, f is convex. [1 mark]

iii) We use Jensen's inequality. Let $f(x) = \max\{x_1, x_2\}$ and $0 \le \theta \le 1$. Then

$$\begin{aligned} f(\theta x + (1-\theta)y) &= \max\{\theta x_1 + (1-\theta)y_1, \theta x_2 + (1-\theta)y_2\} \\ &\le \max\{\theta x_1, \theta x_2\} + \max\{(1-\theta)y_1, (1-\theta)y_2\} \\ &= \theta \max\{x_1, x_2\} + (1-\theta)\max\{y_1, y_2\} \\ &= \theta f(x) + (1-\theta) f(y). \end{aligned}$$

[1 mark]

iv) We use the second-order condition.

$$f(x) = \log(e^{x_1} + e^{x_2}), \quad \nabla f(x) = \begin{bmatrix} \dfrac{e^{x_1}}{e^{x_1} + e^{x_2}} \\ \dfrac{e^{x_2}}{e^{x_1} + e^{x_2}} \end{bmatrix},$$

$$\nabla^2 f(x) = \frac{e^{x_1+x_2}}{(e^{x_1} + e^{x_2})^2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

The term $\dfrac{e^{x_1+x_2}}{(e^{x_1} + e^{x_2})^2}$ is always positive. The matrix is positive semidefinite. This can be shown in several ways, e.g. with the test on the minors (first 1 and second 0), computing the eigenvalues (2 and 0) or showing that the matrix can be written as $\begin{bmatrix} -1 & 1 \end{bmatrix}^\top \begin{bmatrix} -1 & 1 \end{bmatrix}$. Since $\nabla^2 f(x) \ge 0$ for all $x \in \mathbb{R}^2$, f is convex.

1 mark for the Hessian, 1 mark for the conclusion. [2 marks]

b) i) Let $f(x) = ax + b$. As a function of x, $yx - ax - b$ is bounded if and only if $y = a$, in which case it is constant. Therefore,

$$f^*(y) = \begin{cases} -b, & y = a, \\ +\infty, & \text{otherwise.} \end{cases}$$

[2 marks]

ii) Let $f(x) = -\log x$, with $x \in \mathbb{R}_{++}$. The function $xy + \log x$ is unbounded above if $y \ge 0$. For $y < 0$, computing the gradient, the maximum is reached at $x = -1/y$. Therefore,

$$f^*(y) = \begin{cases} -\log(-y) - 1, & y < 0, \\ +\infty, & y \ge 0. \end{cases}$$

[2 marks]

Answers

c) i) Let $f(x) = x\log x$ on $\mathbb{R}_{++}$. Then

$$f'(x) = 1+\log x, \quad f''(x) = \frac{1}{x}, \quad f'''(x) = -\frac{1}{x^2}.$$

Thus

$$\frac{|f'''(x)|}{f''(x)^{\frac{3}{2}}} = \frac{\frac{1}{x^2}}{\frac{1}{x^{\frac{3}{2}}}} = \frac{1}{\sqrt{x}}.$$

Note that as $x \to 0^+$, this is unbounded. Thus, for x small enough we have that $\frac{1}{\sqrt{x}} \leq 2$ fails. This is the case, for instance, for $x = \frac{1}{5}$.

1 mark for the formula, 1 mark for the conclusion. [2 marks]

ii) Let $g(x) = x\log x - \log x$ on $\mathbb{R}_{++}$. Then

$$g'(x) = -\frac{1}{x}+1+\log x, \quad g''(x) = \frac{1}{x^2}+\frac{1}{x}, \quad g'''(x) = -\frac{2}{x^3}-\frac{1}{x^2}.$$

Thus

$$h(x) = \frac{|g'''(x)|}{g''(x)^{\frac{3}{2}}} = \frac{2+x}{(1+x)^{\frac{3}{2}}}.$$

Note that $h(0) = 2$ and $h'(x) = -\dfrac{2+\frac{x}{2}}{(1+x)^{\frac{5}{2}}}$. Recall that we are on $\mathbb{R}_{++}$, thus this last function is always negative. In summary, $h(0) = 2$ and $h(x)$ is strictly decreasing, showing that

$$\frac{|g'''(x)|}{g''(x)^{\frac{3}{2}}} \leq 2.$$

Hence, the function is self-concordant on $\mathbb{R}_{++}$.

1 mark for h, 1 mark for h', 1 mark for the conclusion. [3 marks]

d) i) The centrality equation is given by

$$0 = t\nabla f_0(x^*(t)) + \nabla\phi(x^*(t)) + A^\top \hat{\nu}.$$

Differentiating the centrality equation yields

$$\nabla f_0(z) + \left(t\nabla^2 f_0(z) + \nabla^2\phi(z)\right)\frac{dx^*}{dt} = 0.$$

Thus, the tangent to the central path at $x^*(t)$ is given by

$$\frac{dx^*}{dt} = -\left(t\nabla^2 f_0(x^*(t)) + \nabla^2\phi(x^*(t))\right)^{-1}\nabla f_0(x^*(t)).$$

[2 marks]

ii) To show that $f_0(x^*(t))$ decreases as t increases, we show that

$$\frac{df_0(x^*(t))}{dt} < 0.$$

In fact,

$$\begin{aligned}\frac{df_0(x^*(t))}{dt} &= \nabla f_0(x^*(t))^\top \frac{dx^*}{dt} \\ &= -\nabla f_0(x^*(t))^\top \left(t\nabla^2 f_0(x^*(t)) + \nabla^2\phi(x^*(t))\right)^{-1}\nabla f_0(x^*(t)) < 0,\end{aligned}$$

since it is a negative quadratic form. [2 marks]

Answers

iii) We compute $\hat{x}$ as

$$\hat{x} = x^*(t) - (\mu - 1)t \left(t\nabla^2 f_0(x^*(t)) + \nabla^2 \phi(x^*(t))\right)^{-1} \nabla f_0(x^*(t)).$$

Comparing this with x^n, we notice that the difference is in the terms $t\nabla^2 f_0(x^*(t))$ (for the predictor-corrector method) vs $\mu t \nabla^2 f_0(x^*(t))$ (for Newton's method). Computing x^n and $\hat{x}$ for $f_0(x) = c^\top x$ yields

$$\hat{x} = x^n = x^*(t) - (\mu - 1)t \left(\nabla^2 \phi(x^*(t))\right)^{-1} c,$$

i.e. they are the same. It follows that whenever $\nabla^2 f_0(x^*(t)) = 0$ (e.g. for LPs and QPs), the two methods are identical. [2 marks]

Answers